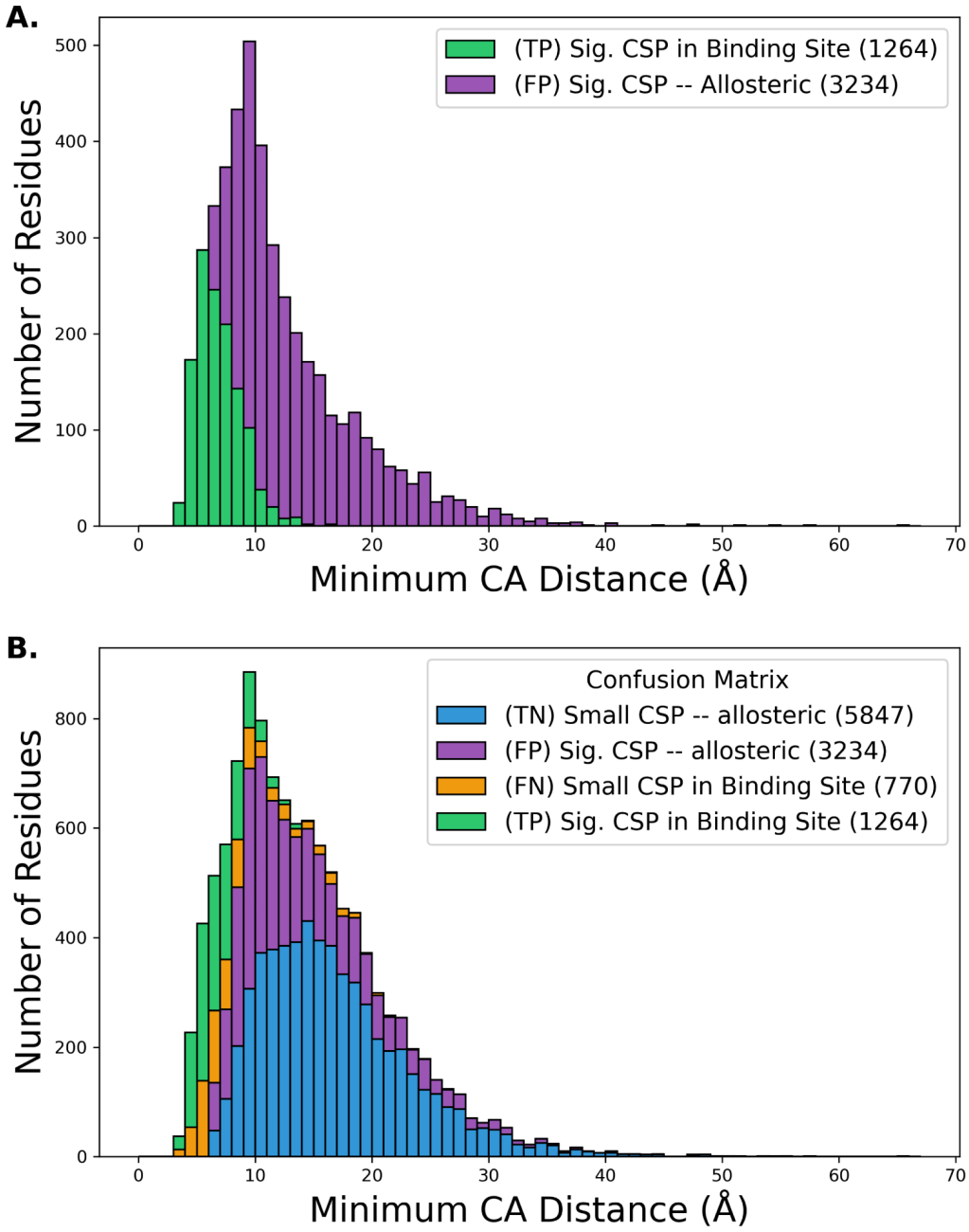
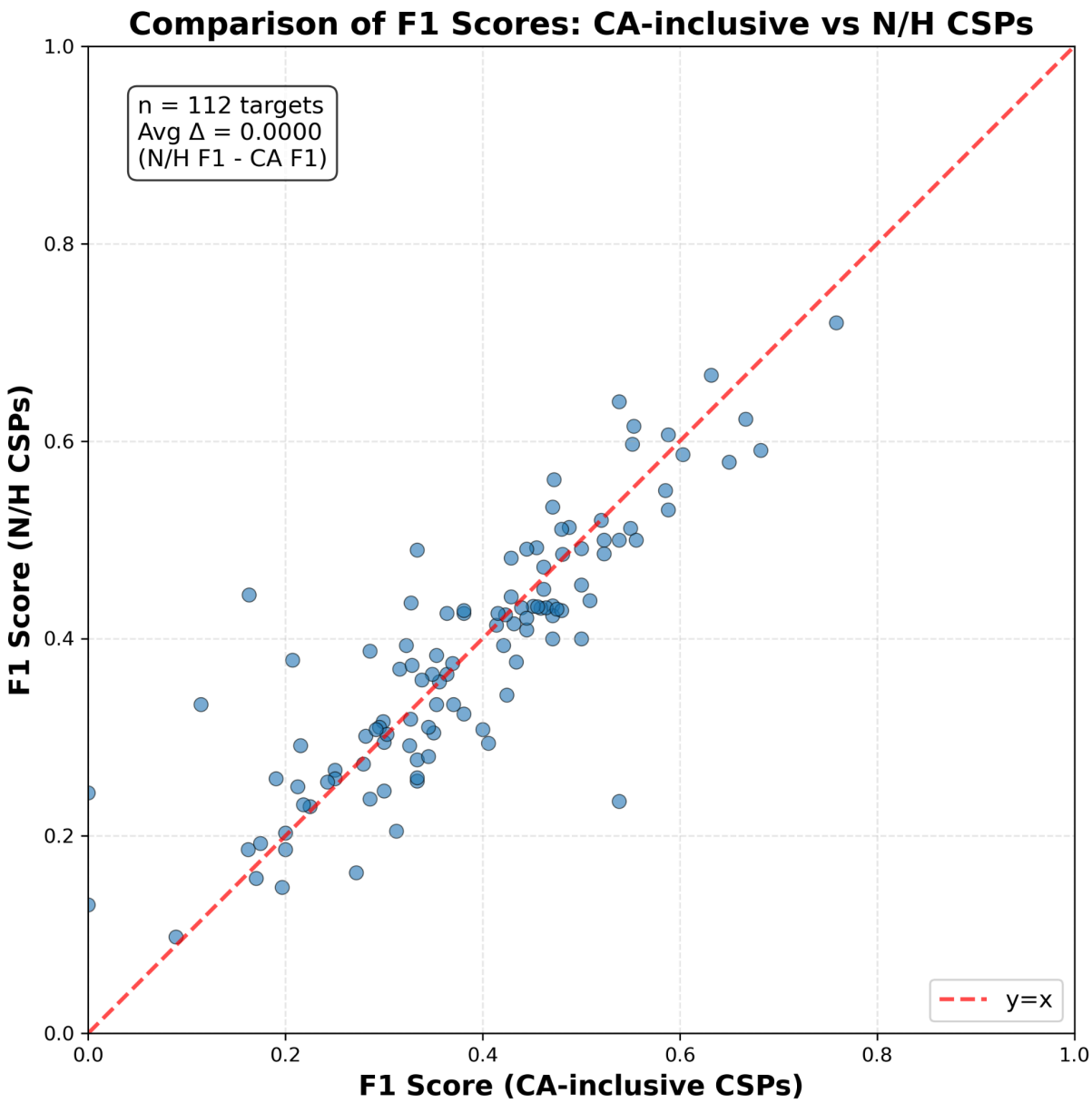
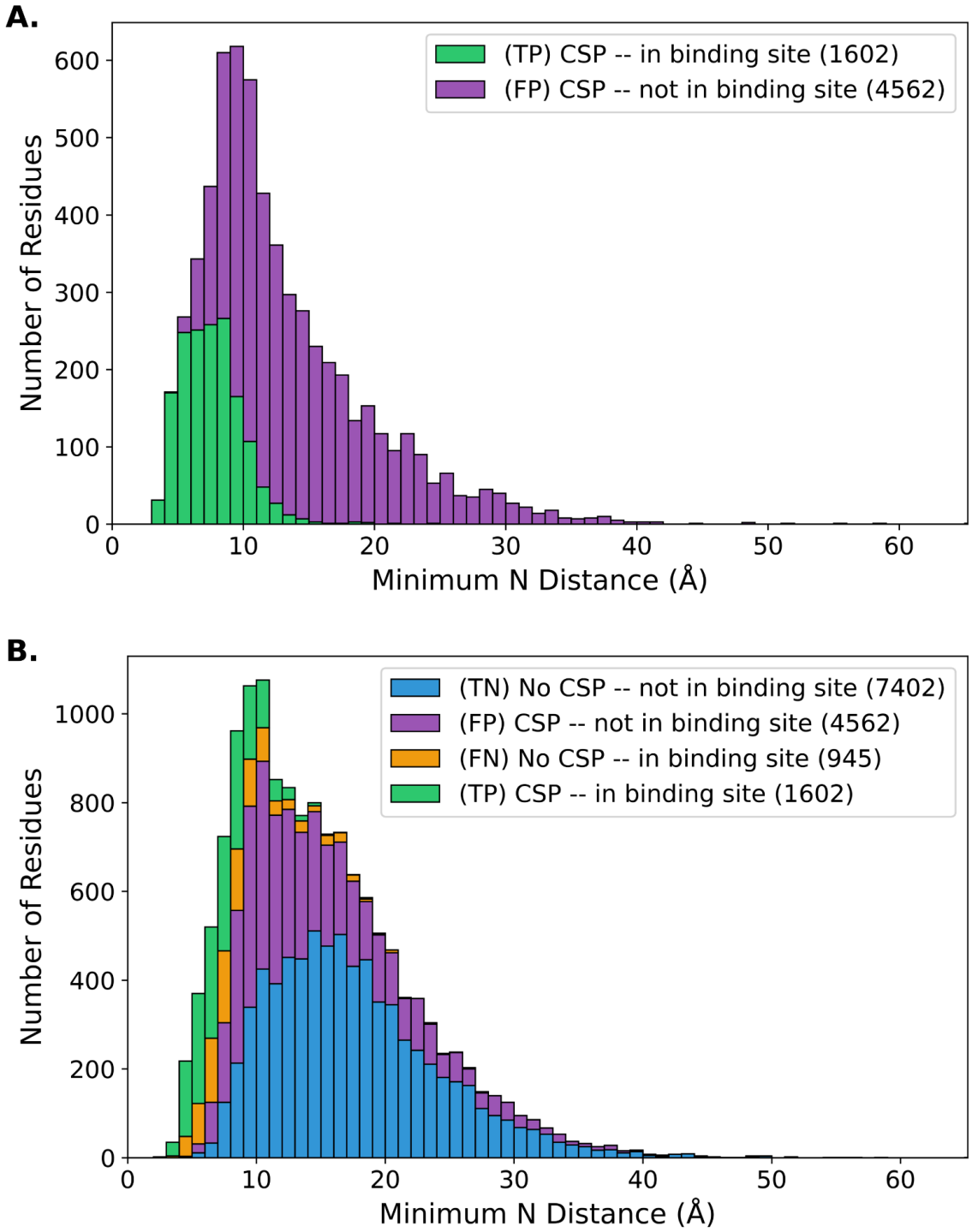


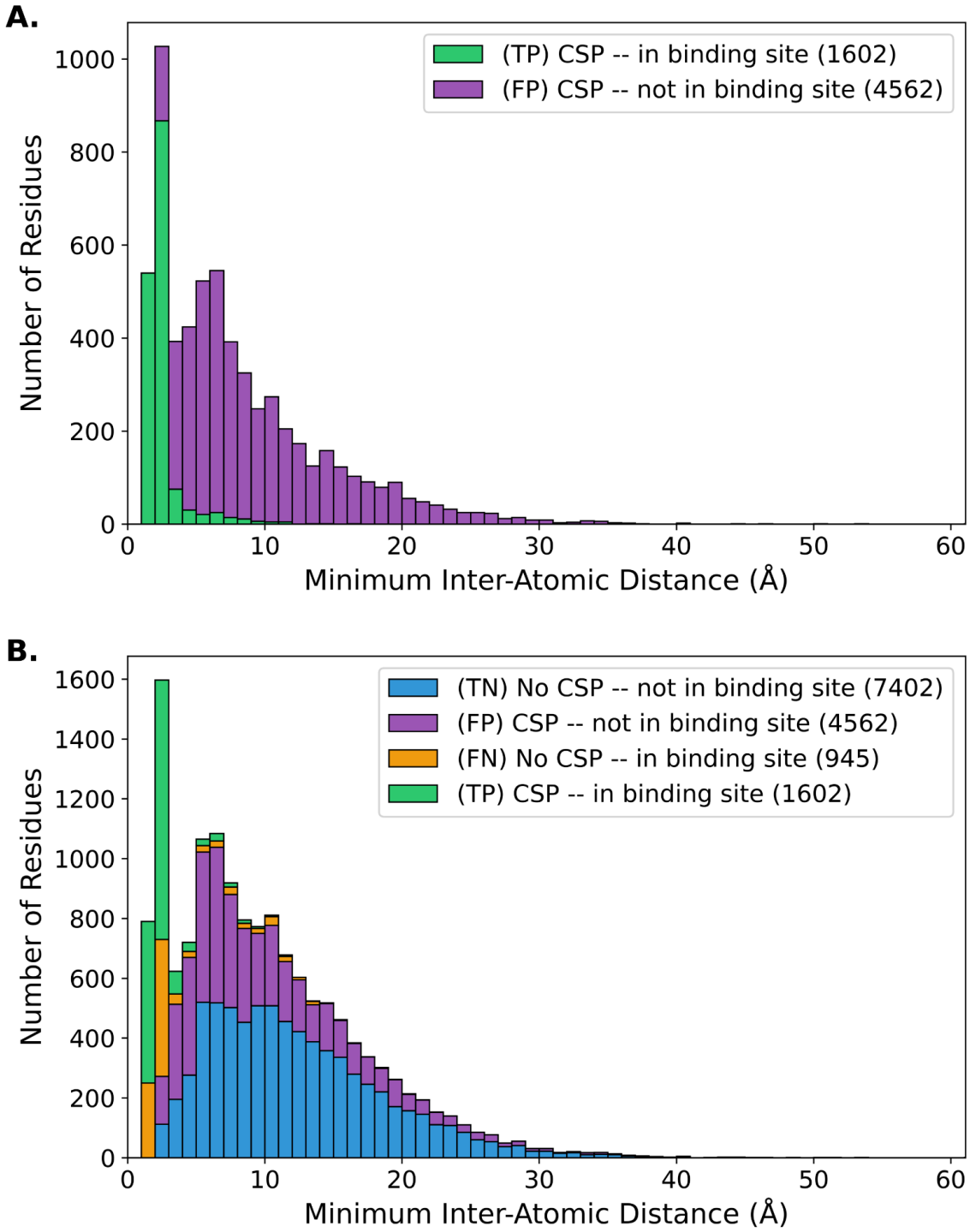
SI Fig. S14. **CA-inclusive CSP Confusion Matrix Histograms**. Calculating CSPs with CAs included in the formula does drastically not change the percentage of allosteric CSPs ~72%. CSPs are calculated using formula described in **SI Eqn. 2**.



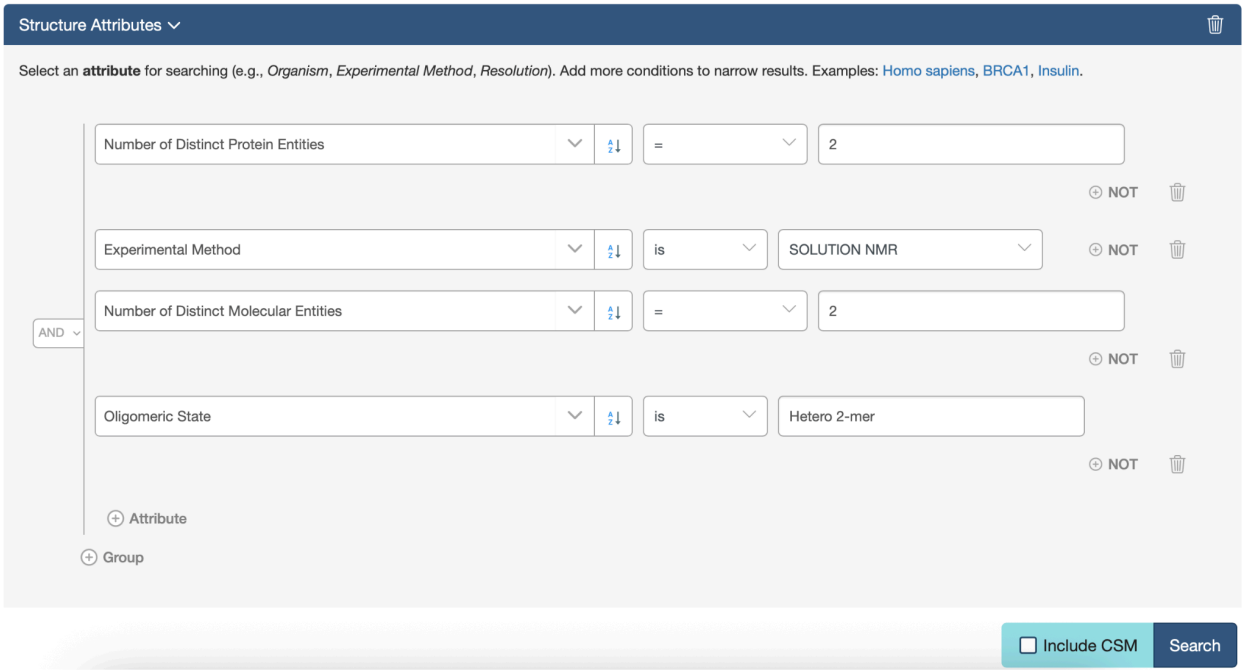
SI Fig. S15. Change in F1 scores when considering CA-inclusive CSPs. This plot shows that over the targets in the CSP DB, the average change in F1 score is negligible when comparing CSPs calculated from N/H shifts vs CSPs calculated with N/H/Ca shifts. CA-inclusive CSPs are calculated following **SI Eqn. 2**.



SI Fig. S16. CSP DB Confusion Matrix Histograms by closest interchain N-N distance.



SI Fig. S17. CSP DB Confusion Matrix Histograms by closest interchain atom-atom distance.



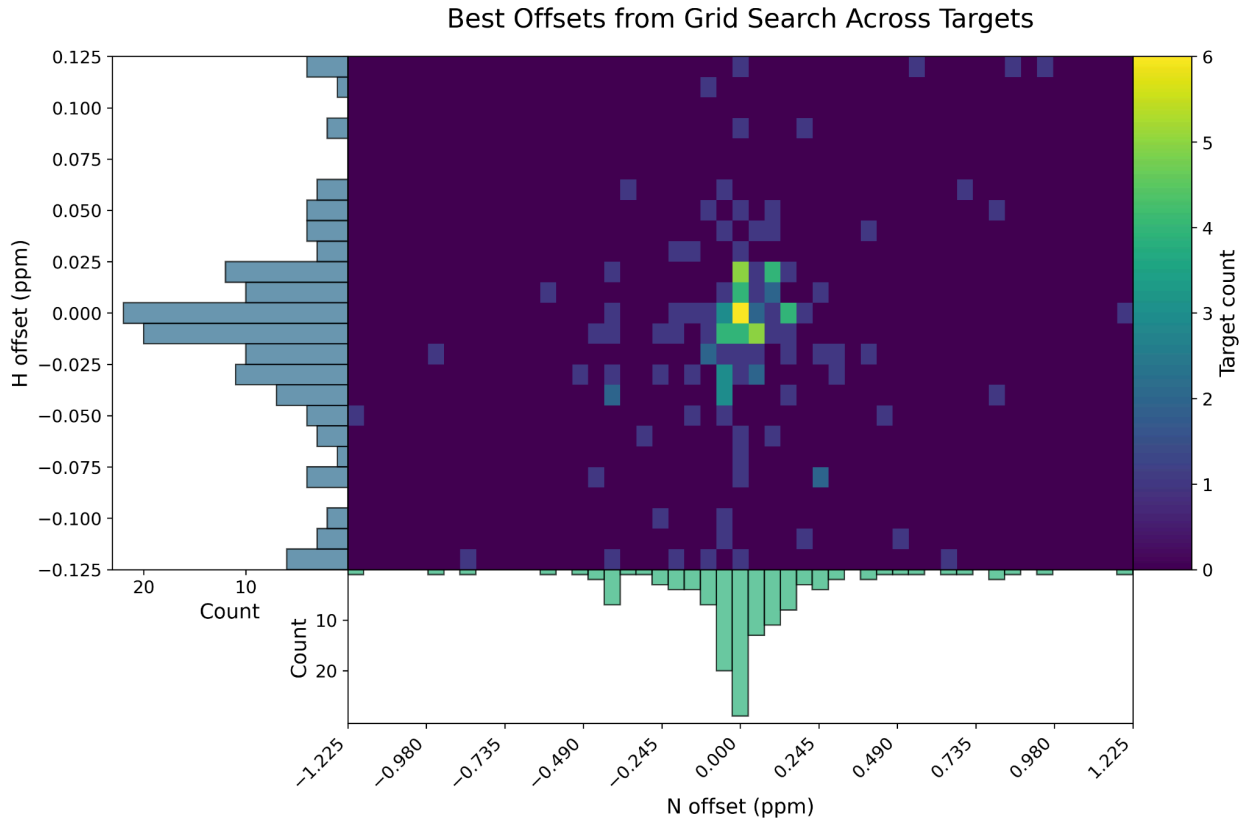
SI Fig. S18. PDB Advanced Search Settings. For the subset of targets manually scraped from the PDB, we restrict our analysis to these settings: entries resolved by Solution NMR, with two and only two distinct protein chains. At the time of curation, this advanced search resulted in 720 hits on the RCSB PDB. From this list of 720, entries were included in the CSP DB if there were N and H chemical shifts deposited in the BMRB for an identical sequence which was evaluated in both apo and holo experiments.

$$\mathrm{CSP} = \sqrt{\frac{1}{2}\left(\Delta\delta_{\mathrm{H}}^2 + (0.14\,\Delta\delta_{\mathrm{N}})^2\right)}$$

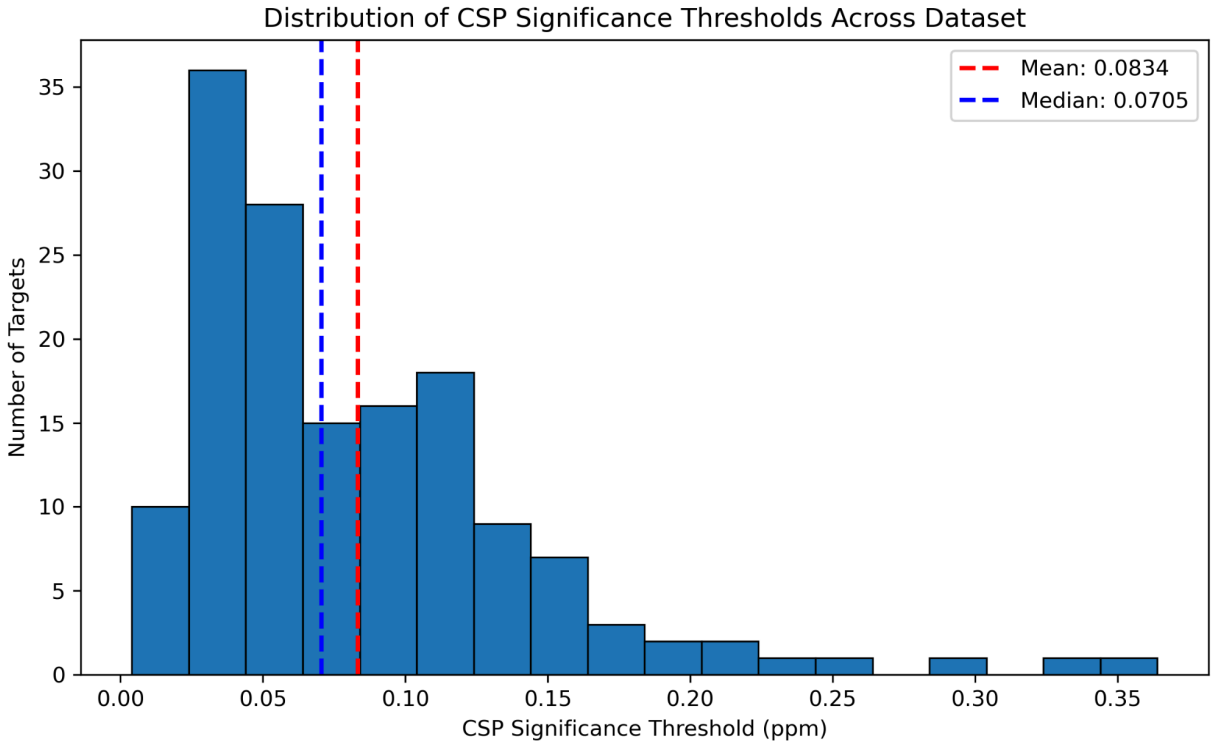
SI Eqn. S1. N/H CSP Equation (Vranken et al., 2005)

$$\mathrm{CSP} = \sqrt{\frac{1}{3}\left(\Delta\delta_{\mathrm{H}}^2 + (0.14\,\Delta\delta_{\mathrm{N}})^2 + (0.3\,\Delta\delta_{\mathrm{C\alpha}})^2\right)}$$

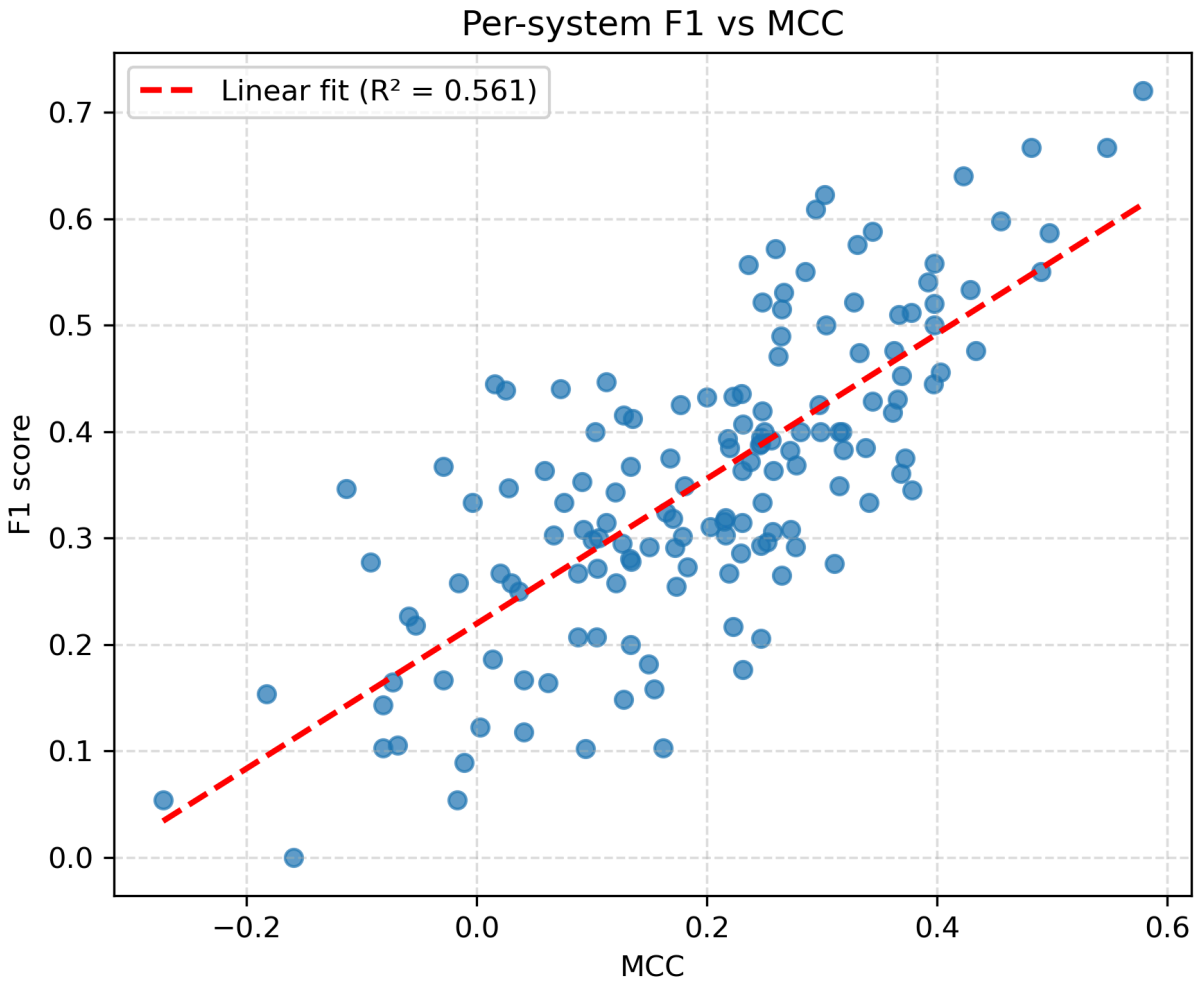
SI Eqn. S2. N/H/Ca CSP Equation (Vranken et al., 2005)



SI Fig. S19. **Ideal offsets across the whole dataset for N/H**. This heatmap shows the offsets used to align the apo/holo HSQC plots. The majority of targets require small adjustments within 0.025 ppm on the Hydrogen axis and 0.245 ppm on the Nitrogen axis.



SI Fig. S20. CSP significance threshold Histogram. This histogram summarizes the thresholds for determining whether a CSP for a given receptor is significant. For a small number of targets, the calculated significance threshold is significantly higher than one would expect, we attribute this to differences in experimental conditions between the apo and holo NMR experiments which aberrantly perturbs all observed shifts. We prune the targets with a calculated significance threshold > 0.6 ppm. This includes the following targets: 2LMC, 1OO9, 2L12, 2L00, 2N01, 2MP0, 8VOI, 8U2Y, 6OSW.



SI Fig. S21. F1 vs MCC score comparison across CSP DB targets. This plot shows the correlation between calculated F1 and MCC metrics. We believe MCC is a more reliable metric for larger systems with many True Negatives (small CSPs outside of the binding site) (Chicco et al., 2021).

References

Agrata, R., & Komander, D. (2025). Ubiquitin—A structural perspective. *Molecular Cell*, *85*(2), 323–346. https://doi.org/10.1016/j.molcel.2024.12.015

Alvarez, G., Sebastian, E., Mondal, A., Roth, M. J., Montelione, G. T., & Perez, A. (2025). *Loop Plasticity Drives Paralog-Specific Recognition in BET ET Domains*. Biophysics. https://doi.org/10.1101/2025.11.11.687848

Chicco, D., Warrens, M. J., & Jurman, G. (2021). The Matthews Correlation Coefficient (MCC) is More Informative Than Cohen's Kappa and Brier Score in Binary Classification Assessment. *IEEE Access*, *9*, 78368–78381. https://doi.org/10.1109/ACCESS.2021.3084050

Gianni, S., Walma, T., Arcovito, A., Calosci, N., Bellelli, A., Engström, Å., Travaglini-Allocatelli, C., Brunori, M., Jemth, P., & Vuister, G. W. (2006). Demonstration of Long-Range Interactions in a PDZ Domain by NMR, Kinetics, and Protein Engineering. *Structure*, *14*(12), 1801–1809. https://doi.org/10.1016/j.str.2006.10.010

Hacisuleyman, A., & Erman, B. (2017). Entropy Transfer between Residue Pairs and Allostery in Proteins: Quantifying Allosteric Communication in Ubiquitin. *PLOS Computational Biology*, *13*(1), e1005319. https://doi.org/10.1371/journal.pcbi.1005319

Juranic, N., Atanasova, E., Filoteo, A. G., Macura, S., Prendergast, F. G., Penniston, J. T., & Strehler, E. E. (2010). Calmodulin Wraps around Its Binding Domain in the Plasma Membrane Ca2+ Pump Anchored by a Novel 18-1 Motif. *Journal of Biological Chemistry*, *285*(6), 4015–4024. https://doi.org/10.1074/jbc.M109.060491

Mondal, A., Swapna, G. V. T., Lopez, M. M., Klang, L., Hao, J., Ma, L., Roth, M. J., Montelione, G. T., & Perez, A. (2023). Structure Determination of Challenging Protein–Peptide Complexes Combining NMR Chemical Shift Data and Molecular Dynamics Simulations.

Journal of Chemical Information and Modeling, *63*(7), 2058–2072. https://doi.org/10.1021/acs.jcim.2c01595

Vranken, W. F., Boucher, W., Stevens, T. J., Fogh, R. H., Pajon, A., Llinas, M., Ulrich, E. L., Markley, J. L., Ionides, J., & Laue, E. D. (2005). The CCPN data model for NMR spectroscopy: Development of a software pipeline. *Proteins: Structure, Function, and Bioinformatics*, *59*(4), 687–696. https://doi.org/10.1002/prot.20449